

PROMYS - Quivers and their representations

Week 1, sheet 1

Problem 1

Two-dimensional vectors. Let $\mathbf{R}^2$ be the set of pairs of real numbers, we will denote such a pair by (a, b) where a, b are real numbers.

- (a) Let V be the set of arrows, or *vectors*, which start at $(0, 0)$ and end somewhere in the plane. Write down a bijection between V and $\mathbf{R}^2$.

So, every pair (a, b) uniquely corresponds to an element of the set V (a vector), which we will denote (in this problem) by $\langle a, b \rangle \in V$.

- (b) For $\lambda \in \mathbf{R}$ a real number, define the map

$$\lambda \cdot : V \rightarrow V$$

by $\lambda \cdot \langle a, b \rangle = \langle \lambda a, \lambda b \rangle$. Geometrically, describe the effect of the map $\lambda \cdot$ acting on the vector $\langle a, b \rangle \in V$.

- (c) Consider a pair of vectors $\langle a, b \rangle \in V$ and $\langle c, d \rangle \in V$. Define the new element

$$\langle a, b \rangle + \langle c, d \rangle = \langle a + c, b + d \rangle.$$

Describe, geometrically, the relationship between the three vectors:

$$\langle a, b \rangle, \quad \langle c, d \rangle, \quad \langle a + c, b + d \rangle.$$

- (d) Simplify the following expression

$$\lambda \cdot (\langle a, b \rangle + \langle c, d \rangle)$$

by writing it as a vector of the form $\langle e, f \rangle$ for some real numbers e, f that can be expressed in terms of λ, a, b, c, d .

- (e) Consider the vectors $\langle 1, 2 \rangle$ and $\langle 2, 1 \rangle$. Find all numbers λ, μ such that:

$$\lambda \langle 1, 2 \rangle + \mu \langle 2, 1 \rangle = \langle 0, 0 \rangle.$$

Remark. Here is a remark on notation. In this problem we wrote an element $v \in V$ as a pair $v = \langle a, b \rangle$. Notationally, this is sometimes written as

$$v = \begin{bmatrix} a \\ b \end{bmatrix}.$$

We refer to this form as a “column vector”.

Problem 2

Linear combinations. Let V be as in the previous problem. A *linear combination* of vectors $v_1, v_2, \dots, v_m \in V$ is a vector of the form

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_m v_m \tag{1}$$

where $\lambda_1, \lambda_2, \dots, \lambda_m$ are real numbers (called the *coefficients*).

- (a) Solve the following system of equations by any means necessary:

$$\begin{aligned} 3x + 2y &= -2 \\ -x + y &= 4 \end{aligned}$$

- (b) Consider the vectors $v_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. For which real numbers λ_1, λ_2 (as in equation (1)) is the vector $\begin{bmatrix} -2 \\ 4 \end{bmatrix}$ a linear combination of v_1, v_2 ?

- (c) Let v_1, v_2 be as in part (2). Find coefficients λ_1, λ_2 such that $\lambda_1 v_1 + \lambda_2 v_2 = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$.

Problem 3

Solving systems of equations Consider the 2×2 system of equations from problem (2.1). The data of this equation can be summarized in the following table:

$$\left[\begin{array}{cc|c} 3 & 2 & -2 \\ -1 & 1 & 4 \end{array} \right] \quad (2)$$

This table is called an *augmented matrix*. Consider the following "row" operations which lead to the solution of this system of equations.

- (1) Multiply row two by 3 to obtain the new augmented matrix.

$$\left[\begin{array}{cc|c} 3 & 2 & -2 \\ -3 & 3 & 12 \end{array} \right] \quad (3)$$

- (2) Add row one to row two to obtain.

$$\left[\begin{array}{cc|c} 3 & 2 & -2 \\ 0 & 5 & 10 \end{array} \right] \quad (4)$$

- (3) Divide row two by 5 to obtain

$$\left[\begin{array}{cc|c} 3 & 2 & -2 \\ 0 & 1 & 2 \end{array} \right] \quad (5)$$

- (4) Subtract 2 times row two from row one to obtain

$$\left[\begin{array}{cc|c} 3 & 0 & -6 \\ 0 & 1 & 2 \end{array} \right] \quad (6)$$

- (5) Finally, divide row one by 3 to obtain

$$\left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 2 \end{array} \right] \quad (7)$$

Translating this back to the variables x, y we see that $x = -2, y = 2$ —the expected answer.

- (a) Perform similar operations to solve the system of equations

$$\begin{aligned} -3x + y &= 4 \\ x + 2y &= 1 \end{aligned}$$

- (b) Set up an "augmented matrix" and perform analogous steps to solve the following 3×3 system of equations (three equations and three unknowns)

$$-3x + 2y - 6z = 6$$

$$5x + 7y - 5z = 6$$

$$x + 4y - 2z = 8.$$

- (c) Can you perform similar row operations to solve the following system of equations

$$x - y = 0$$

$$x + z = 1 \quad ?$$

Can you describe the solution(s) to the above system of equations?

Problem 4

Subspaces. Recall the definition of a subspace of a vector space from lecture.

- (a) Let $V = \mathbf{R}[x]$ be the vector space of polynomials in one variable (whose coefficients are real numbers). Consider the set S of polynomials with nonzero constant term. Is S a subspace of $\mathbf{R}[x]$? Explain why or why not. Let S' be the set of all polynomials with vanishing constant term. Is S' a subspace? Explain why or why not.
- (b) Let $V = \mathbf{R}^2$ be a two-dimensional real vector space. Find subspaces W, W' of $\mathbf{R}^2$ with the property that $W \cup W'$ is *not* a subspace of $\mathbf{R}^2$.
- (c) This problem refers to the system of equations in problem 3(c). Describe the set of *all* solutions to this equation. Does it define a subspace of $\mathbf{R}^3$?
- (d) Show that a homogenous system of linear equations in n -variables $x_1, \dots, x_n$ defines a subspace of $\mathbf{R}^n$. (This fact is independent of the number of equations in the system.)

Problem 5

Challenge problems

- (a) Do there exist polynomials p, q, r, s such that

$$1 + xy + x^2y^2 = p(x)q(y) + r(x)s(y) \quad ?$$

- (b) Let H denote the set of points in $\mathbf{R}^n$ whose coordinates are 0 or 1. Show that a k -dimensional subspace of $\mathbf{R}^n$ can contain at most 2^k points in H . Describe such a subspace.